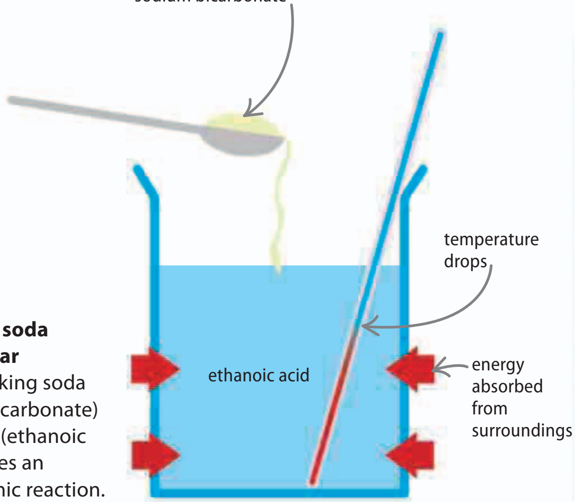
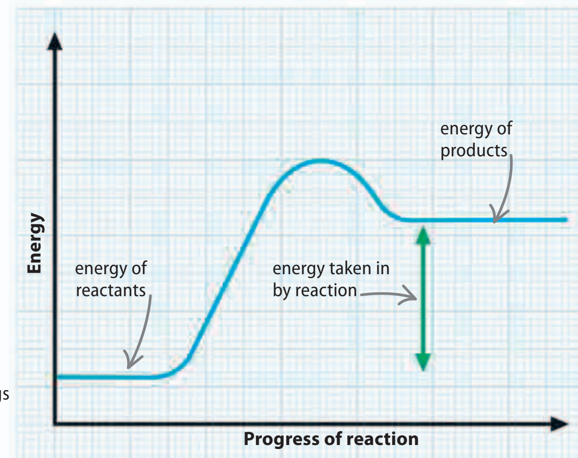
Endothermic reaction

When the amount of energy released during a reaction is less than the activation energy, the process is decribed as endothermic. Because more energy is going into the reaction than is coming out, the reaction mixture and its surroundings become colder as their energy is taken in by the reaction.

▽ **Energy taken in**
In an endothermic reaction the products have more energy than the reactants. This is because energy is taken in from the surroundings during the reaction.

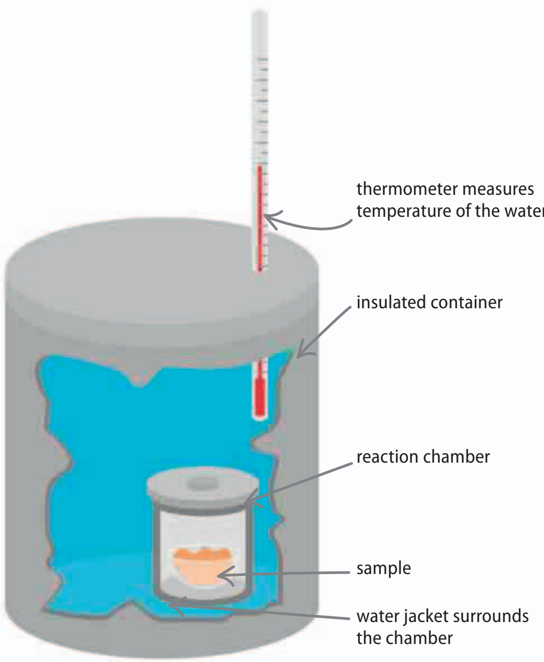


▷ **Baking soda and vinegar**
Adding baking soda (sodium bicarbonate) to vinegar (ethanoic acid) creates an endothermic reaction.



Calorimeter

All the energy used during a chemical reaction can be measured using a calorimeter. A reaction takes place in a central chamber, which is surrounded by water. The calorimeter is completely cut off from the outside, so any changes (rises and falls) in the water temperature can only be a result of the reaction taking place.



▷ **Bomb calorimeter**
This device is used to measure the energy in substances, including different foods. The sample is burned in pure oxygen and the amount of energy released is proportional to the rise in water temperature.

REAL WORLD

Hand warmers

Exothermic reactions are a convenient way of producing heat. Hand-warming packets and self-cooking cans contain two reactants in separate containers. When the hand warmer is bent in half, the containers rupture, mixing the reactants. Their reaction produces harmless products and enough heat to keep hands warm on cold days.